

# Spectral gaps of Cayley graphs of $B_n$ generated by all transpositions and a single odd sign flip

## Abstract

We construct infinite families of Cayley graphs of the Weyl group of type  $B$ , where the generating set is a non-negative symmetric linear combination of diagonal flips with odd cardinality and transpositions. We show that, for these graphs, the random walk and the interchange process have different spectral gaps. This construction contradicts a conjecture of Cesi, which is a generalisation of Aldous' spectral gap conjecture.

*Keywords:* Spectral gap, Weyl group, irreducible representation, interchange process, Aldous' conjecture.

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## 1 Introduction

The second largest eigenvalue of graphs has attracted considerable attention over the past three decades or more [4, 5, 13, 16, 24]. One of the most noteworthy results in this area is *Aldous' spectral gap conjecture* posed by D. Aldous [1] in 1992. It compares two continuous-time Markov chains on a finite connected graph  $G$ : the vertex random walk (RW) and the edge-driven interchange process (IP). If  $L_{\text{RW}}$  and  $L_{\text{IP}}$  denote their generators and throughout we always list eigenvalues as  $0 = \mu_1 < \mu_2 \leq \mu_3 \leq \dots$ , then the conjecture (now a theorem) states

$$\text{gap}(L_{\text{IP}}) = \mu_2(L_{\text{IP}}) = \mu_2(L_{\text{RW}}) = \text{gap}(L_{\text{RW}}). \quad (1.1)$$

Thus, the two chains, despite living on different state spaces, have the same exponential rate of convergence to equilibrium.

Besides the probabilistic interpretation (1.1), Aldous' conjecture can also be examined through the lens of algebraic graph theory, and it becomes evident that it revolves around the algebraic connectivity of Cayley graphs on  $\mathfrak{S}_n$ .

Let  $T \subset \mathfrak{S}_n$  be a generating set of transpositions. Write  $\text{Tra}(T)$  for the transposition graph on  $[n]$  with edge  $\{i, j\}$  if and only if  $(ij) \in T$ , and  $\text{Cay}(\mathfrak{S}_n, T)$  for the Cayley graph of  $\mathfrak{S}_n$  with set of generators  $T$ . For a (regular) graph  $\Gamma$  let  $A(\Gamma)$  be its adjacency matrix and  $L(\Gamma)$  its Laplacian, for  $d$ -regular  $\Gamma$ ,  $L(\Gamma) = dI - A(\Gamma)$ . We can now express Aldous' spectral gap conjecture in an alternative equivalent form

$$\mu_2(L(\text{Cay}(\mathfrak{S}_n, T))) = \mu_2(L(\text{Tra}(T))). \quad (1.2)$$

The representation theory of symmetric groups provides key insights into the essence of Aldous's spectral gap conjecture from an algebraic perspective in graph theory.

Let  $\mathbf{T}^{(n-1,1)}$  denote the standard irreducible representation of  $\mathfrak{S}_n$ . Ordering adjacency eigenvalues in nonincreasing order  $\lambda_1(A) \geq \lambda_2(A) \geq \dots$ , one has

$$\lambda_2(A(\text{Cay}(\mathfrak{S}_n, T))) = \lambda_1(\mathbf{T}^{(n-1,1)}(T)). \quad (1.3)$$

that is, the second largest eigenvalue of  $\text{Cay}(\mathfrak{S}_n, T)$  is achieved by the standard representation of  $\mathfrak{S}_n$ .

After several attempts of special cases [6, 11, 15, 17], Aldous' conjecture was confirmed by Caputo, Liggett, and Richthammer [8] in 2010. Beyond Cayley graphs generated by transpositions, several **exceptional families** remain of active interest. This raises two basic problems: One is to determine, or at least to be sharply bound, the spectral gap; one is to decide whether an Aldous-type conjecture for the gap persists. Illustrative examples include hypergraph analogues [2], initial reversal graphs [10], Cayley graphs of substring-reversal graphs [14], generalised pancake graphs [18], the (extended/complete) alternating group graphs [20], Cayley graphs generated by cycles [21], Cayley graphs built from normal subsets [25].

One important generalisation is Cesi's work [12], he used a map  $\vartheta$  together with the octopus inequality [8, 9] to show a family generating set of the signed symmetric group  $\mathfrak{B}_n$  (See more details in Section 3) that satisfies the Aldous-type conjecture. Let  $[n] := \{1, \dots, n\}$  and, for  $i \in [n]$ , let  $s_{\{i\}} \in \mathfrak{B}_n$  denote the coordinate sign-flip at  $i$ :

$$s_{\{i\}} := \text{diag}(1, \dots, 1, \underbrace{-1}_{i\text{-th entry}}, 1, \dots, 1).$$

**Theorem 1.1** ([12, Thm. 1.2]). *Let  $a_i, b_{ij} \in \mathbb{R}_{\geq 0}$  and*

$$w = \sum_{i=1}^n a_i s_{\{i\}} + \sum_{1 \leq i < j \leq n} b_{ij} (ij), \quad (1.4)$$

*Then the second largest adjacency eigenvalue of  $\text{Cay}(\mathfrak{B}_n, w)$  is achieved by the representation  $\mathbf{P}_n$  of  $\mathfrak{B}_n$ .*

In the final section of his work, Cesi [12] wants to generalise the construction (1.4) to a more general picture. For instance, define

$$Y_n^+ := \{A \subseteq [n] : |A| \equiv 0 \pmod{2}\}, \quad Y_n^- := \{A \subseteq [n] : |A| \equiv 1 \pmod{2}\}.$$

For coefficients  $a_A, b_{ij} \in \mathbb{R}_{\geq 0}$  set  $w_N^\pm := \sum_{A \in Y_n^\pm} a_A s_A$ , and  $w_T := \sum_{(ij) \in T_n} b_{ij} (ij)$ , where  $s_A \in \mathfrak{B}_n$  denotes the diagonal sign-flip with entries  $-1$  on  $A$  and  $+1$  on  $A^c$ . He also showed that, for such  $w := w_N^+ + w_N^- + w_T$ , the Aldous-type property does not always hold. Hence, he made the following conjecture.

**Conjecture 1** ([12, Conj. 5.2]). *Let*

$$w = w_N^- + w_T := \sum_{\substack{A \subseteq [n] \\ |A| \text{ odd}}} a_A s_A + \sum_{1 \leq i < j \leq n} b_{ij} (ij), \quad a_A, b_{ij} \in \mathbb{R}_{\geq 0}. \quad (1.5)$$

*Then the Aldous-type identity (1.1) holds.*

This question also appears in the survey of Liu and Zhou [23], listed as *Problem 33* among the 51 problems collected in the concluding section titled **Open problems**.

In this paper, we disprove Cesi's conjecture 1 by exhibiting families of Cayley graphs (Theorem 5.1) which contradict the identity predicted by the conjecture.

*Remark 1.2 (Related work).* After the completion of this manuscript, we became aware of the independent preprint of Alon and Ghosh [3], arXiv:2510.04244. They study the spectral gap for random walks on  $\mathfrak{B}_n$  generated by transpositions together with arbitrary diagonal sign-flips  $s_A$ , and give a representation-theoretic description of the irreducibles from which the spectral gap may arise and they also show that Cesi's Conjecture 1 is false.

Our approaches are different, and our results are complementary in that we exhibit a particularly sparse and explicit family of counterexamples (one odd flip together with all transpositions) and compute the relevant eigenvalues in closed form, yielding a transparent strict inequality between the Cayley graph spectral gap and the  $2n$ -dimensional  $\mathbf{P}_n$ -block predicted by the conjecture.

## 2 Representations of finite groups

### 2.1 Left regular representations

Let  $G$  be a finite group and let  $w = \sum_{g \in G} w_g g \in \mathbb{C}G$  be symmetric and nonnegative, i.e.  $w_{g^{-1}} = w_g \geq 0$ . Denote by  $\text{Cay}(G, w)$  the (weighted) Cayley graph with vertex set  $G$  and undirected edge  $\{g, hg\}$  carrying weight  $w_h$ . The Laplacian of  $\text{Cay}(G, w)$  acts on functions  $f : G \rightarrow \mathbb{C}$  by

$$L(\text{Cay}(G, w))f(g) = \sum_{h \in G} w_h (f(g) - f(hg)). \quad (2.1)$$

Since  $\text{Cay}(G, w)$  is regular, one also has

$$L(\text{Cay}(G, w)) = \left( \sum_{g \in G} w_g \right) I_N - A(\text{Cay}(G, w)),$$

where  $N = |G|$  and  $A(\text{Cay}(G, w))$  is the weighted adjacency matrix. The least eigenvalue is 0 with a constant eigenvector. We define the spectral gap of the Cayley graph by

$$\text{gap}(\text{Cay}(G, w)) := \mu_2(L(\text{Cay}(G, w))).$$

Let  $\mathbf{L}$  be the left-regular representation of  $G$  on  $\mathbb{C}^G$ ,  $(\mathbf{L}(h)f)(g) = f(h^{-1}g)$ . Using the symmetry of  $w$ ,

$$L(\text{Cay}(G, w)) = \sum_{h \in G} w_h (I_N - \mathbf{L}(h)). \quad (2.2)$$

The formula (2.2) motivates the following representation-theoretic extension. For a (complex) representation  $R : G \rightarrow \text{GL}(V)$ , set

$$L_G(R, w) := \sum_{h \in G} w_h (I_V - R(h)). \quad (2.3)$$

Let  $t := \dim V^{G, R}$  be the multiplicity of the trivial representation in  $R$ . Enumerating eigenvalues of  $L_G(R, w)$  as  $\mu_1 \leq \dots \leq \mu_{\dim V}$ , the spectral gap of the pair  $(w, R)$  is

$$\text{gap}_G(R, w) := \mu_{t+1}(L_G(R, w)),$$

with the convention that  $\mu_{t+1} = +\infty$  if  $t = \dim V$ . The Cayley case is the special instance  $R = \mathbf{L}$ :

$$L(\text{Cay}(G, w)) = L_G(\mathbf{L}, w), \quad \text{gap}(\text{Cay}(G, w)) = \text{gap}_G(\mathbf{L}, w). \quad (2.4)$$

Although  $\mathbf{L}$  contains all irreducibles, in many situations one can identify a small family of representations that already determines the gap:

$$\text{gap}_G(\mathbf{L}, w) = \min_{T \in \mathcal{F}} \text{gap}_G(\mathbf{T}, w) \quad \text{for a suitable finite set } \mathcal{F} \subset \text{Irr}(G).$$

This replacement dramatically reduces matrix size while preserving the value of the gap.

## 2.2 The spectral gap of the representation Laplacian

Let  $G$  be a finite group. Write  $\text{rep}(G)$  for the class of all finite-dimensional complex representations of  $G$ , and  $\text{Irr}(G)$  for the set of equivalence classes of irreducibles. By Maschke's theorem, every  $\mathbf{R} \in \text{rep}(G)$  decomposes as

$$\mathbf{R} \cong \bigoplus_{\mathbf{T} \in \text{Irr}(G)} \text{mult}(\mathbf{R}, \mathbf{T}) \mathbf{T}, \quad (2.5)$$

where  $\text{mult}(\mathbf{R}, \mathbf{T}) \in \mathbb{Z}_{\geq 0}$  is the multiplicity of  $\mathbf{T}$  in  $\mathbf{R}$ . If  $\mathbf{R}$  acts on the complex vector space  $V$ , the subspace of  $G$ -invariant vectors is

$$V_G^{\mathbf{R}} := \{ v \in V : \mathbf{R}(g)v = v \text{ for all } g \in G \}.$$

By definition,

$$\dim V_G^{\mathbf{R}} = \text{mult}(\mathbf{R}, \mathbf{1}), \quad (2.6)$$

where  $\mathbf{1}$  denotes the one dimensional trivial representation.

Let  $\mathbb{C}G$  be the group algebra. For  $w = \sum_{g \in G} w_g g \in \mathbb{C}G$  set

$$\text{supp } w := \{ g \in G : w_g \neq 0 \}, \quad w^* := \sum_{g \in G} \overline{w_g} g^{-1}.$$

We call  $w$  symmetric if  $w = w^*$  and positive if  $w_g \geq 0$  for all  $g \in G$ . Put

$$\mathbb{C}G^{(s)} := \{ w \in \mathbb{C}G : w = w^* \}, \quad \mathbb{R}_+^{(s)} := \{ w \in \mathbb{C}G : w = w^* \text{ and } w_g \geq 0 \ \forall g \}.$$

For a pair  $(\mathbf{R}, w)$ , we write  $L_G(\mathbf{R}, w)$  for the associated (representation) Laplacian. If  $w$  is symmetric,  $L_G(\mathbf{R}, w)$  is Hermitian and hence has real eigenvalues; if  $w \in \mathbb{R}_+^{(s)}$ , these eigenvalues are nonnegative. Enumerate them in nondecreasing order (with multiplicity) as

$$\mu_1(L_G(\mathbf{R}, w)) \leq \mu_2(L_G(\mathbf{R}, w)) \leq \cdots \leq \mu_s(L_G(\mathbf{R}, w)),$$

where  $s = \deg \mathbf{R} = \dim V$ . An eigenvalue is trivial if its eigenspace is contained in  $V_G^{\mathbf{R}}$ . With the convention  $\min \emptyset = +\infty$ , define the spectral gap of the pair  $(\mathbf{R}, w)$  by

$$\text{gap}_G(\mathbf{R}, w) := \min\{ \mu \in \text{Spec } L_G(\mathbf{R}, w) : \mu \text{ is nontrivial} \}. \quad (2.7)$$

If  $t := \dim V_G^{\mathbf{R}} = \text{mult}(\mathbf{R}, \mathbf{1})$ , then  $L_G(\mathbf{R}, w)$  has exactly  $t$  trivial eigenvalues (counted with multiplicity), and therefore

$$\text{gap}_G(\mathbf{R}, w) = \mu_{t+1}(L_G(\mathbf{R}, w)). \quad (2.8)$$

Minimising over representations yields the spectral gap of  $w$ :

$$\text{gap}_G(w) := \inf_{\mathbf{R} \in \text{rep}(G)} \text{gap}_G(\mathbf{R}, w). \quad (2.9)$$

From (2.5) we have

$$\text{spec } L_G(\mathbf{R}, w) = \bigcup_{\substack{\mathbf{T} \in \text{Irr}(G) \\ \text{mult}(\mathbf{R}, \mathbf{T}) > 0}} \text{spec } L_G(\mathbf{T}, w). \quad (2.10)$$

whence

$$\text{gap}_G(\mathbf{R}, w) = \min_{\substack{\mathbf{T} \in \text{Irr}(G) \\ \text{mult}(\mathbf{R}, \mathbf{T}) > 0}} \text{gap}_G(\mathbf{T}, w), \quad \text{and} \quad \text{gap}_G(w) = \min_{\mathbf{T} \in \text{Irr}(G)} \text{gap}_G(\mathbf{T}, w). \quad (2.11)$$

Let  $\mathbf{L}$  be the left-regular representation of  $G$ . Since

$$\mathbf{L} \cong \bigoplus_{\mathbf{T} \in \text{Irr}(G)} (\deg \mathbf{T}) \mathbf{T} \quad \text{and} \quad \text{mult}(\mathbf{L}, \mathbf{1}) = 1,$$

we obtain

$$\text{gap}_G(\mathbf{L}, w) = \mu_2(L_G(\mathbf{L}, w)). \quad (2.12)$$

Thus, the above notions recover the usual interpretation of the spectral gap of the (weighted) Cayley graph  $\text{Cay}(G, w)$  as its second smallest eigenvalue.

### 3 Weyl groups $\mathfrak{B}_n$

Let  $\mathfrak{B}_n$  denote the Weyl group of type  $B_n$  (also called the hyperoctahedral group, or the signed symmetric group). We shall work with two equivalent realisations and move freely between them.

**Realisations of  $\mathfrak{B}_n$**  We can understand  $\mathfrak{B}_n$  in a matrix way, that is,

$$\mathfrak{B}_n = \left\{ M \in \text{GL}(n, \mathbb{C}) : \begin{array}{l} \text{each row and column of } M \text{ has exactly one nonzero entry, equal to } \pm 1 \end{array} \right\}.$$

Under this realisation, the symmetric group  $\mathfrak{S}_n$  is embedded as a subgroup of positive monomial matrices (all nonzero entries  $+1$ ). Let

$$N_n := \{\text{diag}(\varepsilon_1, \dots, \varepsilon_n) : \varepsilon_i \in \{\pm 1\}\} \cong (\mathbb{Z}/2\mathbb{Z})^n.$$

Then  $N_n \triangleleft \mathfrak{B}_n$  and  $\mathfrak{B}_n = N_n \rtimes \mathfrak{S}_n$ , so  $|\mathfrak{B}_n| = 2^n n!$ .

Equivalently,

$$\mathfrak{B}_n \cong (\mathbb{Z}/2\mathbb{Z})^n \rtimes \mathfrak{S}_n,$$

the semidirect product where  $\mathfrak{S}_n$  permutes the  $n$  sign coordinates. In particular  $\mathfrak{B}_n$  acts faithfully on the signed set

$$X_n = \{-n, \dots, -1, 1, \dots, n\},$$

by  $(\eta, \pi) \cdot k := \text{sgn}(k)(-1)^{\eta_{\pi(|k|)}}\pi(|k|)$ , and hence admits a faithful permutation representation

$$\mathbf{P}_n : \mathfrak{B}_n \hookrightarrow \mathfrak{S}_{2n}, \quad \text{Im}(\mathbf{P}_n) = \{\sigma \in \mathfrak{S}_{2n} : \sigma(-k) = -\sigma(k) \ \forall k\}.$$

Here,  $\mathbf{P}_n$  is the  $2n$ -dimensional block that will control spectral gaps later, in fact (see Proposition 3.1)  $\mathbf{P}_n$  splits as the direct sum of the  $n$ -dimensional defining representation of  $\mathfrak{B}_n$  and the pullback of the defining representation of  $\mathfrak{S}_n$ .

**Irreducible representations and (bi)partitions** A **partition** of  $n$  is a nonincreasing finite sequence  $\alpha = (\alpha_1, \dots, \alpha_r)$  of positive integers with  $\sum_{i=1}^r \alpha_i = n$ . The **size** is  $|\alpha| := \sum_{i=1}^r \alpha_i$ ; we write  $\alpha \vdash n$  when  $|\alpha| = n$ . Irreducible representations of  $\mathfrak{S}_n$  are indexed (up to equivalence) by partitions of  $n$ . If  $\alpha \vdash n$ , the corresponding irreducible is denoted by  $[\alpha]$ . For branching, we use the standard sets

$$\begin{aligned} \alpha \downarrow &:= \{\gamma : \gamma \text{ is obtained from } \alpha \text{ by removing one box}\}, \\ \alpha \uparrow &:= \{\beta : \beta \text{ is obtained by adding one box}\}. \end{aligned}$$

We also reserve the notation  $\mu_1(M) \leq \mu_2(M) \leq \dots$  for the ordered eigenvalues of a Hermitian matrix  $M$ , and write  $\text{gap}(\cdot)$  for spectral gaps when they appear later.

Follow Cesi's notation [12], we view  $\mathfrak{B}_n$  as the set of pairs  $(\eta, \pi)$  with  $\eta \in \{0, 1\}^n$  and  $\pi \in \mathfrak{S}_n$ , equipped with the product

$$(\eta, \pi) \cdot (\zeta, \sigma) := (\eta + \zeta \circ \pi^{-1} \pmod{2}, \pi\sigma), \quad (3.1)$$

where the sum on  $\{0, 1\}^n$  is componentwise modulo 2 and  $(\zeta \circ \pi^{-1})_i := \zeta_{\pi^{-1}(i)}$ . Set

$$N_n := \{(\eta, \text{id}) : \eta \in \{0, 1\}^n\} \cong (\mathbb{Z}/2\mathbb{Z})^n, \quad S_n := \{(0, \pi) : \pi \in \mathfrak{S}_n\} \cong \mathfrak{S}_n.$$

Then  $N_n \triangleleft \mathfrak{B}_n$  and  $\mathfrak{B}_n = N_n \rtimes S_n$ . We shall freely identify  $S_n$  with  $\mathfrak{S}_n$  via  $\pi \mapsto (0, \pi)$ .

As a natural generalisation, irreducible representations of  $\mathfrak{B}_n$  are parametrised by **bipartitions**  $(\alpha, \beta)$  with  $|\alpha| + |\beta| = n$ . We denote the corresponding irreducible representations by  $[\alpha, \beta]$ . Fix  $k := |\alpha|$  and  $n - k := |\beta|$ . Let

$$\mathfrak{B}_{n,k} := N_n \rtimes (\mathfrak{S}_k \times \mathfrak{S}_{n-k}) \leq \mathfrak{B}_n,$$

where  $\mathfrak{S}_k \times \mathfrak{S}_{n-k}$  acts by permuting the first  $k$  and the last  $n - k$  coordinates. Define the 1-dimensional character  $U_k$  of  $N_n$  by

$$U_k(\eta) := (-1)^{\#\{i \in \{k+1, \dots, n\} : \eta_i = 1\}}, \quad \eta \in \{0, 1\}^n, \quad (3.2)$$

and extend it trivially across  $\mathfrak{S}_k \times \mathfrak{S}_{n-k}$ . Set

$$\Theta_{\alpha, \beta} := U_k \boxtimes [\alpha] \boxtimes [\beta] \quad \text{as a representation of } \mathfrak{B}_{n,k}.$$

Then

$$[\alpha, \beta] \cong \text{Ind}_{\mathfrak{B}_{n,k}}^{\mathfrak{B}_n} \Theta_{\alpha, \beta}, \quad (3.3)$$

and every irrep of  $\mathfrak{B}_n$  arises uniquely in this way (up to equivalence). In particular, the trivial representation corresponds to  $[(n), \emptyset]$ , and the determinant character of  $\mathfrak{B}_n$  corresponds to  $[\emptyset, (1^n)]$ .

**Branching rule** For the symmetric group the classical branching reads

$$[\alpha] \downarrow_{\mathfrak{S}_{n-1}} \cong \bigoplus_{\gamma \in \alpha \downarrow} [\gamma], \quad [\gamma] \uparrow^{\mathfrak{S}_n} \cong \bigoplus_{\beta \in \gamma \uparrow} [\beta]. \quad (3.4)$$

The analogue for  $\mathfrak{B}_n$  is obtained by removing one box either from the first or from the second component:

$$[\alpha, \beta] \downarrow_{\mathfrak{B}_{n-1}} \cong \left( \bigoplus_{\gamma \in \alpha \downarrow} [\gamma, \beta] \right) \oplus \left( \bigoplus_{\delta \in \beta \downarrow} [\alpha, \delta] \right). \quad (3.5)$$

**Defining representations** Let  $\mathbf{D}_n^\circ$  denote the defining  $n$ -dimensional representation of  $\mathfrak{S}_n$ , realised by permutation matrices:

$$(\mathbf{D}_n^\circ(\pi))_{ij} = \delta_{i,\pi(j)}, \quad \pi \in \mathfrak{S}_n. \quad (3.6)$$

As is well known,

$$\mathbf{D}_n^\circ \cong [n] \oplus [n-1, 1]. \quad (3.7)$$

For  $\mathfrak{B}_n$  we take the faithful defining representation  $\mathbf{D}_n$  on  $\mathbb{C}^n$ , given on the standard basis  $(e_1, \dots, e_n)$  by

$$\mathbf{D}_n(\eta, \pi) e_j = (-1)^{\eta_{\pi(j)}} e_{\pi(j)}, \quad (\eta, \pi) \in \mathfrak{B}_n. \quad (3.8)$$

Equivalently, the matrix entries are  $(\mathbf{D}_n(\eta, \pi))_{ij} = (-1)^{\eta_i} \delta_{i,\pi(j)}$ . The image of  $\mathfrak{B}_n$  under  $\mathbf{D}_n$  is the group of signed permutation matrices, its restriction to  $\mathfrak{S}_n$  recovers (3.6). Moreover,

$$\mathbf{D}_n \cong [(n-1), (1)], \quad (3.9)$$

hence  $\mathbf{D}_n$  is irreducible of degree  $n$ .

Since  $\mathfrak{B}_n/N_n \cong \mathfrak{S}_n$ , any representation  $\mathbf{R}$  of  $\mathfrak{S}_n$  pulls back to a representation  $\tilde{\mathbf{R}}$  of  $\mathfrak{B}_n$  by

$$\tilde{\mathbf{R}}(\eta, \pi) := \mathbf{R}(\pi), \quad (\eta, \pi) \in \mathfrak{B}_n. \quad (3.10)$$

In particular, the pullback of  $\mathbf{D}_n^\circ$  will be denoted  $\tilde{\mathbf{D}}_n^\circ$ ; its matrix entries are the same as in (3.6). From (3.7) we obtain

$$\tilde{\mathbf{D}}_n^\circ \cong [(n), \emptyset] \oplus [(n-1, 1), \emptyset]. \quad (3.11)$$

Set  $V_n := \mathbb{C}X_n$  with basis  $\{e_x : x \in X_n\}$  and define the associated permutation representation  $\mathbf{P}_n$  by

$$\mathbf{P}_n(\eta, \pi)e_x := e_{(\eta, \pi) \cdot x}.$$

Equivalently, the matrix elements are

$$(\mathbf{P}_n(\eta, \pi))_{xy} = \mathbf{1} \quad \{x = (\eta, \pi) \cdot y\}.$$

The representation  $\mathbf{P}_n$  is faithful and has degree  $2n$ . The following decomposition will be used later.

**Proposition 3.1** ([12, Prop. 3.1]). We have a  $\mathfrak{B}_n$ -equivariant decomposition

$$\mathbf{P}_n \cong \mathbf{D}_n \oplus \tilde{\mathbf{D}}_n^\circ \cong [(n-1), (1)] \oplus [(n), \emptyset] \oplus [(n-1, 1), \emptyset]. \quad (3.12)$$

**Corollary 3.2** ([12, Equation 2.7+Proposition 3.1]). *Let  $w \in \mathbb{C}\mathfrak{B}_n$  be symmetric and nonnegative. Then*

$$\text{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w) = \min \left\{ \text{gap}_{\mathfrak{B}_n}(\mathbf{D}_n, w), \text{gap}_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w) \right\}. \quad (3.13)$$

*In particular,*

$$\text{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w) = \min \left\{ \mu_1(L_{\mathfrak{B}_n}(\mathbf{D}_n, w)), \mu_1(L_{\mathfrak{B}_n}([(n-1, 1), \emptyset], w)) \right\}. \quad (3.14)$$

## 4 Cesi's conjecture

**Theorem 4.1.** *Let  $n = 4$  and  $t > 0$ . For*

$$w_t := t s_{\{1,2,3\}} + \sum_{1 \leq i < j \leq 4} (ij) \in \mathbb{C}\mathfrak{B}_4.$$

*Then*

$$\text{gap}_{\mathfrak{B}_4}(w_t) < \text{gap}_{\mathfrak{B}_4}(\mathbf{P}_4, w_t).$$

### 4.1 Value of $\text{gap}_{\mathfrak{B}_4}(\mathbf{P}_4, w_t)$

**Lemma 4.2.** *Write  $w_t = w_N^- + w_T$  with*

$$w_N^- := t s_{\{1,2,3\}}, \quad w_T := \sum_{1 \leq i < j \leq 4} (ij).$$

*Then*

$$L_{\mathfrak{B}_4}(\tilde{\mathbf{D}}_4^\circ, w_N^-) = 0, \quad (4.1)$$

$$L_{\mathfrak{B}_4}(\mathbf{D}_4, w_N^-) = 2 \text{diag}(t, t, t, 0), \quad (4.2)$$

$$L_{\mathfrak{B}_4}(\tilde{\mathbf{D}}_4^\circ, w_T) = L_{\mathfrak{B}_4}(\mathbf{D}_4, w_T). \quad (4.3)$$

*Proof.* By definition,  $L_{\mathfrak{B}_4}(\mathbf{R}, w) = \sum_{h \in \mathfrak{B}_4} w_h (I - \mathbf{R}(h))$ . For the two 4-dimensional models,

$$[\mathbf{D}_4(\eta, \pi)]_{ij} = (-1)^{\eta_i} \delta_{i, \pi(j)}, \quad [\tilde{\mathbf{D}}_4^\circ(\eta, \pi)]_{ij} = \delta_{i, \pi(j)}.$$

Hence, for a diagonal sign  $s_A = (\eta_A, \text{id})$ , we have

$$\tilde{\mathbf{D}}_4^\circ(s_A) = I_4 \quad \text{and} \quad \mathbf{D}_4(s_A) = \text{diag}(x_i), \quad x_i = \begin{cases} -1, & i \in A, \\ +1, & i \notin A. \end{cases}$$

This gives (4.1) and (4.2).

For (4.3), if  $(ij) \in S_4 \subset \mathfrak{B}_4$  then  $\mathbf{D}_4((ij)) = \tilde{\mathbf{D}}_4^\circ((ij))$  (the same permutation matrix), and linearity of  $L$  yields the claim.  $\square$

**$\tilde{\mathbf{D}}_4^\circ$ -block** By (4.1),  $L_{\mathfrak{B}_4}(\tilde{\mathbf{D}}_4^\circ, w_t) = L_{\mathfrak{B}_4}(\tilde{\mathbf{D}}_4^\circ, w_T)$ . Now  $w_T = \sum_{i < j} (ij)$  acts, in the defining  $\mathfrak{S}_4$ -representation, as the Laplacian of the complete graph  $K_4$ , hence

$$\text{gap}_{\mathfrak{B}_4}(\tilde{\mathbf{D}}_4^\circ, w_t) = \mu_2(L(K_4)) = 4$$

as  $L(K_n) = nI - J$  has spectrum  $\{0, n\}$ .

**$\mathbf{D}_4$ -block** By (4.2)–(4.3),

$$L_{\mathfrak{B}_4}(\mathbf{D}_4, w_t) = L(K_4) + 2 \text{diag}(t, t, t, 0).$$

Thus  $L_{\mathfrak{B}_4}(\mathbf{D}_4, w_t)$  equals the symmetric matrix

$$M(t) = \begin{pmatrix} 3 + 2t & -1 & -1 & -1 \\ -1 & 3 + 2t & -1 & -1 \\ -1 & -1 & 3 + 2t & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix},$$

whose spectrum is

$$\text{Spec } M(t) = \{4 + 2t, 4 + 2t, t + 2 \pm \sqrt{(t - 1)^2 + 3}\}.$$

Hence

$$\text{gap}_{\mathfrak{B}_4}(\mathbf{D}_4, w_t) = \mu_{\min}(M(t)) = t + 2 - \sqrt{(t - 1)^2 + 3}.$$

Since  $\mu_{\min}(M(t)) < 4$  for all  $t > 0$ , the smaller block determines the gap of  $\mathbf{P}_4$ :

$$\text{gap}_{\mathfrak{B}_4}(\mathbf{P}_4, w_t) = t + 2 - \sqrt{(t - 1)^2 + 3}. \quad (4.4)$$

## 4.2 An upper bound for $\text{gap}_{\mathfrak{B}_4}(w_t)$

By Equation (2.11), it is quite natural to choose one of the irreducible representations, which must provide an upper bound for  $\text{gap}_{\mathfrak{B}_4}(w_t)$ .

**Realisation of  $[(2), (2)]$**  We realise the irreducible  $[(2), (2)]$  as the induced module

$$[(2), (2)] \cong \text{Ind}_H^{\mathfrak{B}_4}(U_2), \quad H := N_4 \cdot (\mathfrak{S}_2 \times \mathfrak{S}_2),$$

where  $U_2$  is the one-dimensional character of  $N_4 \cong (\mathbb{Z}/2\mathbb{Z})^4$ , extended trivially to  $\mathfrak{S}_2 \times \mathfrak{S}_2$ . Since  $|\mathfrak{B}_4 : H| = 384/64 = 6$ , the induced space has dimension 6.

Choose representatives  $t_S = (0, \pi_S)$  with  $\pi_S \in \mathfrak{S}_4$  such that  $\pi_S(\{1, 2\}) = S$ , where  $S$  is a 2-subsets of  $[4]$ . Via

$$\Phi : \mathfrak{B}_4/H \longrightarrow \binom{[4]}{2}, \quad (\eta, \pi)H \longmapsto \pi(\{1, 2\}),$$

the natural basis  $\{1_{t_S H}\}$  is indexed by 2-subsets  $S \subset [4]$ . Write  $e_S := 1_{t_S H}$  and fix

$$\mathcal{B} = \left\{ \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 4\}, \{2, 4\}, \{3, 4\} \right\}.$$

The action is explicitly given by<sup>1</sup>

$$\mathbf{R}((0, \sigma)) e_S = e_{\sigma(S)}, \quad \mathbf{R}(s_A) e_S = (-1)^{|A \cap S^c|} e_S \quad (s_A \in N_4). \quad (4.5)$$

The first action of (4.5) comes from that, for any  $g, t \in \mathfrak{B}_4$  one has

$$(\mathbf{R}(g) 1_{tH})(x) = 1_{tH}(g^{-1}x),$$

so the support of  $\mathbf{R}(g) 1_{tH}$  is the left coset  $gtH$ , hence  $\mathbf{R}(g) 1_{tH} = 1_{gtH}$ . Taking  $g = (0, \sigma)$  and  $t_S = (0, \pi_S)$  yields

$$\mathbf{R}((0, \sigma)) e_S = \mathbf{R}((0, \sigma)) 1_{t_S H} = 1_{(0, \sigma)t_S H} = 1_{(0, \sigma\pi_S)H} = 1_{t_{\sigma(S)}H} = e_{\sigma(S)},$$

because  $(\sigma\pi_S)(\{1, 2\}) = \sigma(S)$ .

In the left-coset model of  $\mathbf{R} = \text{Ind}_H^{\mathfrak{B}_4}(U_2)$  with the standard normalisation  $1_{tH}(th) = U_2(h)$  for  $h \in H$ , one has

$$\mathbf{R}(h) 1_{tH} = U_2(t^{-1}ht) 1_{tH} \quad (h \in H).$$

Specialising to  $h = s_A = (\eta_A, \text{id})$  and  $t = t_S = (0, \pi_S)$  with  $\pi_S(\{1, 2\}) = S$ , we get  $t_S^{-1} s_A t_S = s_{\pi_S^{-1}(A)}$  and, since  $U_2(s_B) = (-1)^{|B \cap \{3, 4\}|}$ , it follows that

$$\mathbf{R}(s_A) e_S = (-1)^{|\pi_S^{-1}(A) \cap \{3, 4\}|} e_S = (-1)^{|A \cap S^c|} e_S,$$

which is precisely the second identity in (4.5).

---

<sup>1</sup>The following notation  $S^c$  means the complement of  $S$ .

**Transposition part** Among the six transpositions  $(ij)$ , exactly two fix  $S$  (the one exchanging the elements of  $S$  and the one exchanging the elements of  $S^c$ ; the remaining four send  $S$  to the four neighbours of  $S$  in the Johnson graph  $J(4, 2)$ . Hence,

$$\sum_{1 \leq i < j \leq 4} \mathbf{R}((ij)) = 2I + A(J(4, 2)),$$

and therefore

$$L_{\mathfrak{B}_4}(\mathbf{R}, w_T) = \sum_{i < j} (I - \mathbf{R}((ij))) = 4I - A(J(4, 2)). \quad (4.6)$$

Using  $\text{Spec } A(J(4, 2)) = \{4, 0, -2\}$  with multiplicities 1, 3, 2, we obtain

$$\text{Spec } L_{\mathfrak{B}_4}(\mathbf{R}, w_T) = \{0, 4, 4, 4, 6, 6\}.$$

**Odd-flip part.** By (4.5),  $\mathbf{R}(s_{\{1,2,3\}})e_S = (-1)^{|S^c \cap \{1,2,3\}|} e_S$ , thus

$$L_{\mathfrak{B}_4}(\mathbf{R}, w_N^-) = t(I - \mathbf{R}(s_{\{1,2,3\}})) = \text{diag}(2t, 2t, 2t, 0, 0, 0). \quad \text{in the basis } \mathcal{B}. \quad (4.7)$$

Hence, the representation Laplacian is

$$L_{\mathfrak{B}_4}(\mathbf{R}, w_t) = L_{\mathfrak{B}_4}(\mathbf{R}, w_T) + L_{\mathfrak{B}_4}(\mathbf{R}, w_N^-) = (4I - A(J(4, 2))) + \text{diag}(2t, 2t, 2t, 0, 0, 0),$$

namely

$$M_6(t) = \begin{pmatrix} 4 + 2t & -1 & -1 & -1 & -1 & 0 \\ -1 & 4 + 2t & -1 & -1 & 0 & -1 \\ -1 & -1 & 4 + 2t & 0 & -1 & -1 \\ -1 & -1 & 0 & 4 & -1 & -1 \\ -1 & 0 & -1 & -1 & 4 & -1 \\ 0 & -1 & -1 & -1 & -1 & 4 \end{pmatrix}.$$

The spectrum (with algebraic multiplicities) is

$$\text{Spec } M_6(t) = \{[5 + t \pm \sqrt{t^2 + 1}]^2, 2 + t \pm \sqrt{t^2 + 4}\}.$$

and the smallest eigenvalue is

$$\mu_{\min}(L_{\mathfrak{B}_4}(\mathbf{R}, w_t)) = 2 + t - \sqrt{t^2 + 4}.$$

since  $5 + t - \sqrt{t^2 + 1} > 4$  for all  $t > 0$ . Hence

$$\text{gap}_{\mathfrak{B}_4}([(2), (2)], w_t) = 2 + t - \sqrt{t^2 + 4}. \quad (4.8)$$

### 4.3 Comparison

Combining (4.4) with (4.8) and noting that

$$t^2 + 4 - ((t-1)^2 + 3) = 2t > 0 \quad (t > 0),$$

we obtain  $\sqrt{t^2 + 4} > \sqrt{(t-1)^2 + 3}$  and hence

$$2 + t - \sqrt{t^2 + 4} < 2 + t - \sqrt{(t-1)^2 + 3}.$$

Therefore

$$\text{gap}_{\mathfrak{B}_4}(w_t) \leq \text{gap}_{\mathfrak{B}_4}([(2), (2)], w_t) < \text{gap}_{\mathfrak{B}_4}(\mathbf{P}_4, w_t),$$

which yields the desired strict inequality.

For the Cayley graph  $\text{Cay}(\mathfrak{B}_4, w_t)$  one has

$$\text{gap}_{\mathfrak{B}_4}(w_t) < \text{gap}_{\mathfrak{B}_4}(\mathbf{P}_4, w_t) = \min\left\{\text{gap}_{\mathfrak{B}_4}(\mathbf{D}_4, w_t), \text{gap}_{\mathfrak{B}_4}(\tilde{\mathbf{D}}_4^\circ, w_t)\right\}.$$

Hence, the spectral gap of  $\text{Cay}(\mathfrak{B}_4, w_t)$  is not realised by

the standard  $\mathfrak{B}_4$ -representation  $\mathbf{D}_4$  (nor by  $\tilde{\mathbf{D}}_4^\circ$ ). That is to say, the continuous-time random walk and the interchange process on  $\mathfrak{B}_4$  (with generator  $w_t$ ) have different spectral gaps.  $\square$

## 5 A generalisation of the above generating set

In this section, we consider a generalisation of the generating set described above, which consists of an infinite family of Cayley graphs of the Weyl group of type  $B$ . We demonstrate that there are always infinite counterexamples to Cesi's conjecture for any  $n$ .

**Theorem 5.1.** *Let  $n \geq 4$ ,  $t > 0$ , and let  $A \subset [n]$  be any odd subset with  $|A| = s$ . For*

$$w_{t,s} := t s_A + \sum_{1 \leq i < j \leq n} (ij) \in \mathbb{C} \mathfrak{B}_n,$$

*we have, for every  $n - 1 \geq s > \frac{n}{2}$ ,*

$$\text{gap}_{\mathfrak{B}_n}(w_{t,s}) \leq \text{gap}_{\mathfrak{B}_n}([(1), (n-1)], w_{t,s}) < \text{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s}).$$

That is to say, for these generating sets, the spectral gap of the Cayley graph is not realised by the  $2n$ -dimensional permutation representation  $\mathbf{P}_n$ .

## 5.1 Value of $\text{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s})$

**Lemma 5.2.** Write  $w_{t,s} = w_N^- + w_T$  with

$$w_N^- := t s_A, \quad w_T := \sum_{1 \leq i < j \leq n} (ij).$$

Then,

$$L_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_N^-) = 0, \quad (5.1)$$

$$L_{\mathfrak{B}_n}(\mathbf{D}_n, w_N^-) = 2t \text{diag}(\mathbf{1}_A), \quad (5.2)$$

$$L_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_T) = L_{\mathfrak{B}_n}(\mathbf{D}_n, w_T). \quad (5.3)$$

*Proof.* By definition  $L_{\mathfrak{B}_n}(\mathbf{R}, w) = \sum_h w_h (I - \mathbf{R}(h))$ . For the two  $n$ -dimensional models one has

$$[\mathbf{D}_n(\eta, \pi)]_{ij} = (-1)^{\eta_i} \delta_{i, \pi(j)}, \quad [\tilde{\mathbf{D}}_n^\circ(\eta, \pi)]_{ij} = \delta_{i, \pi(j)}.$$

Hence, for a diagonal sign  $s_A$ ,  $\tilde{\mathbf{D}}_n^\circ(s_A) = I_n$  and  $\mathbf{D}_n(s_A) = \text{diag}(x_i)$  with  $x_i = -1$  for  $i \in A$  and  $+1$  otherwise. Therefore

$$L_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_N^-) = t(I - \tilde{\mathbf{D}}_n^\circ(s_A)) = 0,$$

and

$$L_{\mathfrak{B}_n}(\mathbf{D}_n, w_N^-) = t(I - \mathbf{D}_n(s_A)) = t \text{diag}(1 - x_i) = 2t \text{diag}(\mathbf{1}_A).$$

For (5.3), each transposition  $(ij) \in \mathfrak{S}_n \subset \mathfrak{B}_n$  acts as the same permutation matrix in both models, hence linearity of  $L$  yields the claim.  $\square$

**$\tilde{\mathbf{D}}_n^\circ$ -block** By (5.1),  $L_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_{t,s}) = L_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_T)$ . Since  $w_T$  acts, in the defining  $\mathfrak{S}_n$ -representation, as the Laplacian of the complete graph  $K_n$ , we have

$$L_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_{t,s}) = nI - J, \quad \text{Spec} = \{0, n, \dots, n\},$$

hence  $\text{gap}_{\mathfrak{B}_n}(\tilde{\mathbf{D}}_n^\circ, w_{t,s}) = n$ .

**D<sub>n</sub>-block** By (5.2)–(5.3),

$$L_{\mathfrak{B}_n}(\mathbf{D}_n, w_{t,s}) = (nI - J) + 2t \operatorname{diag}(\mathbf{1}_A).$$

whose spectrum is

$$\operatorname{Spec} M(t) = \{ [n]^{n-s-1}, [n+2t]^{s-1}, \frac{n+2t \pm \sqrt{(n+2t)^2 - 8ts}}{2} \}.$$

Therefore, combining the results above, we have

$$\operatorname{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s}) = \operatorname{gap}_{\mathfrak{B}_n}(\mathbf{D}_n, w_{t,s}) = \frac{n+2t - \sqrt{(n+2t)^2 - 8ts}}{2}. \quad (5.4)$$

## 5.2 An upper bound for $\operatorname{gap}_{\mathfrak{B}_n}(w_{t,s})$

By the minimisation over irreducible representations,  $\operatorname{gap}_{\mathfrak{B}_n}(w) \leq \operatorname{gap}_{\mathfrak{B}_n}(\mathbf{R}, w)$  for every representation  $\mathbf{R}$ . We choose the irreducible

$$\mathbf{R}_1 = [(1), (n-1)] \cong \operatorname{Ind}_H^{\mathfrak{B}_n}(U_1), \quad H := N_n \cdot (\mathfrak{S}_1 \times \mathfrak{S}_{n-1}),$$

where  $U_1$  is the standard 1-dimensional character of  $N_n \simeq (\mathbb{Z}/2\mathbb{Z})^n$ . The induced space has dimension  $n$  and can be indexed by  $i \in [n]$ .

**Realisation of  $\mathbf{R}_1$**  In the left-coset representation with basis  $(e_i)_{i=1}^n$ ,

$$\mathbf{R}_1((0, \sigma)) e_i = e_{\sigma(i)}, \quad \mathbf{R}_1(s_A) e_i = (-1)^{|A \setminus \{i\}|} e_i = \begin{cases} +e_i, & i \in A, \\ -e_i, & i \notin A, \end{cases} \quad (5.5)$$

since  $|A|$  is odd.

**Transposition part** As in the defining  $\mathfrak{S}_n$ -representation,  $\sum_{i < j} (ij)$  acts by

$$L_{\mathfrak{B}_n}(\mathbf{R}_1, w_T) = nI - J$$

(the Laplacian of  $K_n$ ), exactly as in  $\mathbf{D}_n$ .

**Odd–flip part** From (5.5),

$$L_{\mathfrak{B}_n}(\mathbf{R}_1, w_N^-) = t(I - \mathbf{R}_1(s_A)) = 2t \operatorname{diag}(\mathbf{1}_{A^c}).$$

Hence

$$L_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s}) = (nI - J) + 2t \operatorname{diag}(\mathbf{1}_{A^c}).$$

whose spectrum is

$$\operatorname{Spec} M_c(t) = \left\{ [n]^{s-1}, [n+2t]^{n-s-1}, \frac{n+2t \pm \sqrt{(n+2t)^2 - 8t(n-s)}}{2} \right\},$$

Therefore

$$\operatorname{gap}_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s}) = \frac{n+2t - \sqrt{(n+2t)^2 - 8t(n-s)}}{2}. \quad (5.6)$$

### 5.3 Comparison

Consider  $\phi(c) := \frac{n+2t - \sqrt{(n+2t)^2 - 4c}}{2}$  on  $(0, \infty)$ . Then

$$\phi'(c) = \frac{1}{\sqrt{(n+2t)^2 - 4c}} > 0,$$

so  $\phi$  is strictly increasing. Equations (5.4)–(5.6) read

$$\operatorname{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s}) = \phi(2ts), \quad \operatorname{gap}_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s}) = \phi(2t(n-s)).$$

If  $s > \frac{n}{2}$ , then  $2t(n-s) < 2ts$ . Hence,

$$\operatorname{gap}_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s}) < \operatorname{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s}).$$

Finally, by the general minimisation identity over irreducible representations,

$$\operatorname{gap}_{\mathfrak{B}_n}(w_{t,s}) \leq \operatorname{gap}_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s}) < \operatorname{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s}),$$

which yields Theorem 5.1. □

*Remark 5.3.* If  $t = 0$  then  $\operatorname{supp}(w_{t,s})$  generates only  $\mathfrak{S}_n$  and  $\operatorname{Cay}(\mathfrak{B}_n, w_{t,s})$  is disconnected, hence  $\operatorname{gap}_{\mathfrak{B}_n}(w_{t,s}) = 0$ .

If  $t > 0$  with  $|A|$  odd with  $1 \leq |A| \leq n-1$ , the group generated by  $\{(ij)\}_{1 \leq i < j \leq n} \cup \{s_A\}$  equals  $\mathfrak{B}_n$ , so the Cayley graph is connected and all gaps above are strictly positive.

If  $|A| = n$  (necessarily  $n$  odd), then  $s_A = -I$  is central and  $\langle \{(ij)\}, s_A \rangle \cong \mathfrak{S}_n \times \langle -I \rangle \subsetneq \mathfrak{B}_n$ . Thus  $\operatorname{Cay}(\mathfrak{B}_n, w_{t,s})$  is disconnected and  $\operatorname{gap}_{\mathfrak{B}_n}(w_{t,s}) = 0$ .

Based on the discussion above, we propose two conjectures.

**Conjecture 2.** *Let  $n \geq 4$ ,  $t > 0$ , and  $A \subset [n]$  be an odd subset with  $s := |A|$  satisfying  $n - 1 \geq s > \frac{n}{2}$ . Set*

$$w_{t,s} := t s_A + \sum_{1 \leq i < j \leq n} (ij) \in \mathbb{C} \mathfrak{B}_n.$$

*Let  $\mathbf{R}_1 = [(1), (n-1)]$  be the  $n$ -dimensional irreducible representation. Then the spectral gap of the Cayley graph  $\text{Cay}(\mathfrak{B}_n, w_{t,s})$  is realised by  $\mathbf{R}_1$ , i.e.*

$$\text{gap}_{\mathfrak{B}_n}(w_{t,s}) = \text{gap}_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s}).$$

*Equivalently, one has the explicit formula*

$$\text{gap}_{\mathfrak{B}_n}(w_{t,s}) = \frac{n + 2t - \sqrt{(n + 2t)^2 - 8t(n - s)}}{2}.$$

**Conjecture 3.** *Let  $n \geq 4$ ,  $t > 0$ , and let  $A \subset [n]$  be an odd subset with  $1 \leq s := |A| \leq n - 1$ . Set*

$$w_{t,s} := t s_A + \sum_{1 \leq i < j \leq n} (ij) \in \mathbb{C} \mathfrak{B}_n.$$

*Define*

$$\phi_{n,t}(c) := \frac{n + 2t - \sqrt{(n + 2t)^2 - 4c}}{2}, \quad 0 \leq c \leq \frac{(n + 2t)^2}{4}.$$

*Then the spectral gap of  $\text{Cay}(\mathfrak{B}_n, w_{t,s})$  is given by*

$$\text{gap}_{\mathfrak{B}_n}(w_{t,s}) = \min\{\text{gap}_{\mathfrak{B}_n}(\mathbf{P}_n, w_{t,s}), \text{gap}_{\mathfrak{B}_n}(\mathbf{R}_1, w_{t,s})\} = \phi_{n,t}(2t \min\{s, n - s\}).$$

*Equivalently,*

$$\text{gap}_{\mathfrak{B}_n}(w_{t,s}) = \min \left\{ \frac{n + 2t - \sqrt{(n + 2t)^2 - 8ts}}{2}, \frac{n + 2t - \sqrt{(n + 2t)^2 - 8t(n - s)}}{2} \right\}.$$

*In particular, if  $s > \frac{n}{2}$  then the gap is realised by  $\mathbf{R}_1 = [(1), (n-1)]$ , whereas if  $s < \frac{n}{2}$  it is realised by the  $\mathbf{P}_n$ -block (equivalently by  $\mathbf{D}_n = [(n-1), (1)]$ ).*

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